SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Dr.V.PARTHIPAN

Code of Conduct:

I Ahamed Kaif S (192321128) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

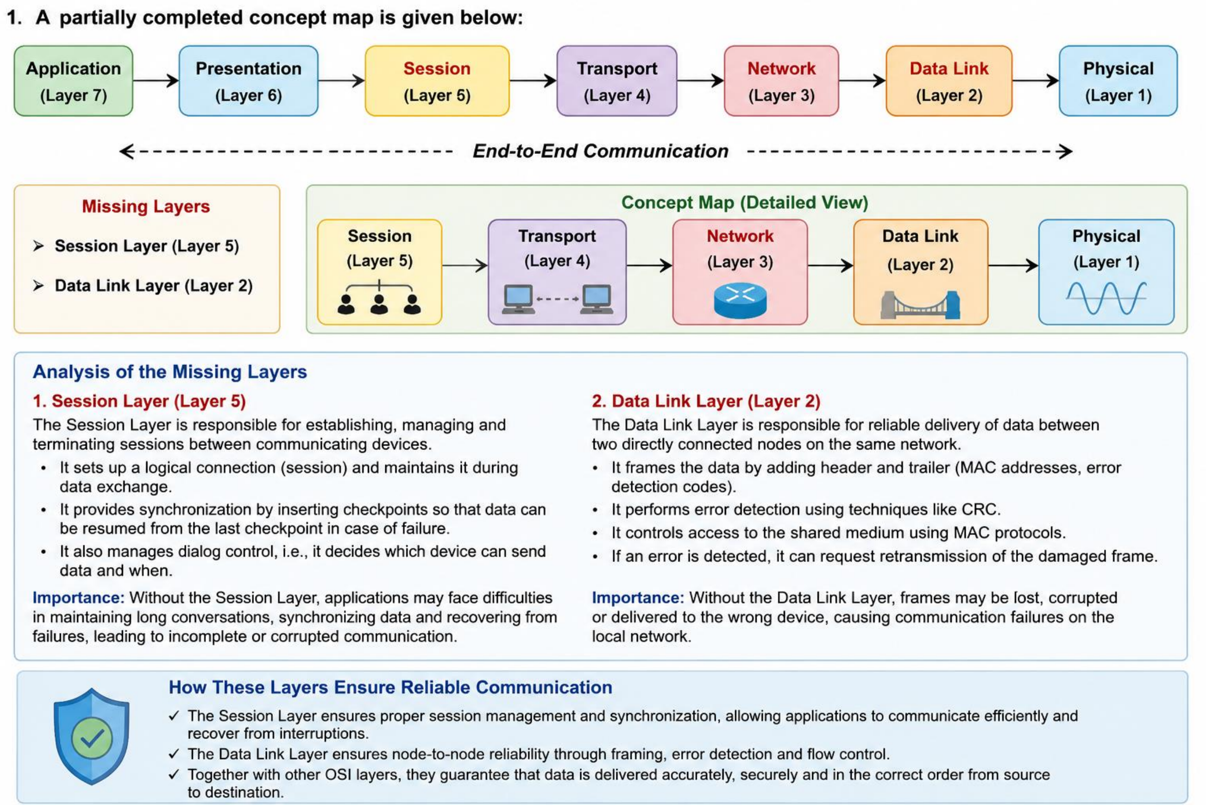
Signature of the student:
Ahamed Kaif S

Annexure A
CONCEPT MAPPING

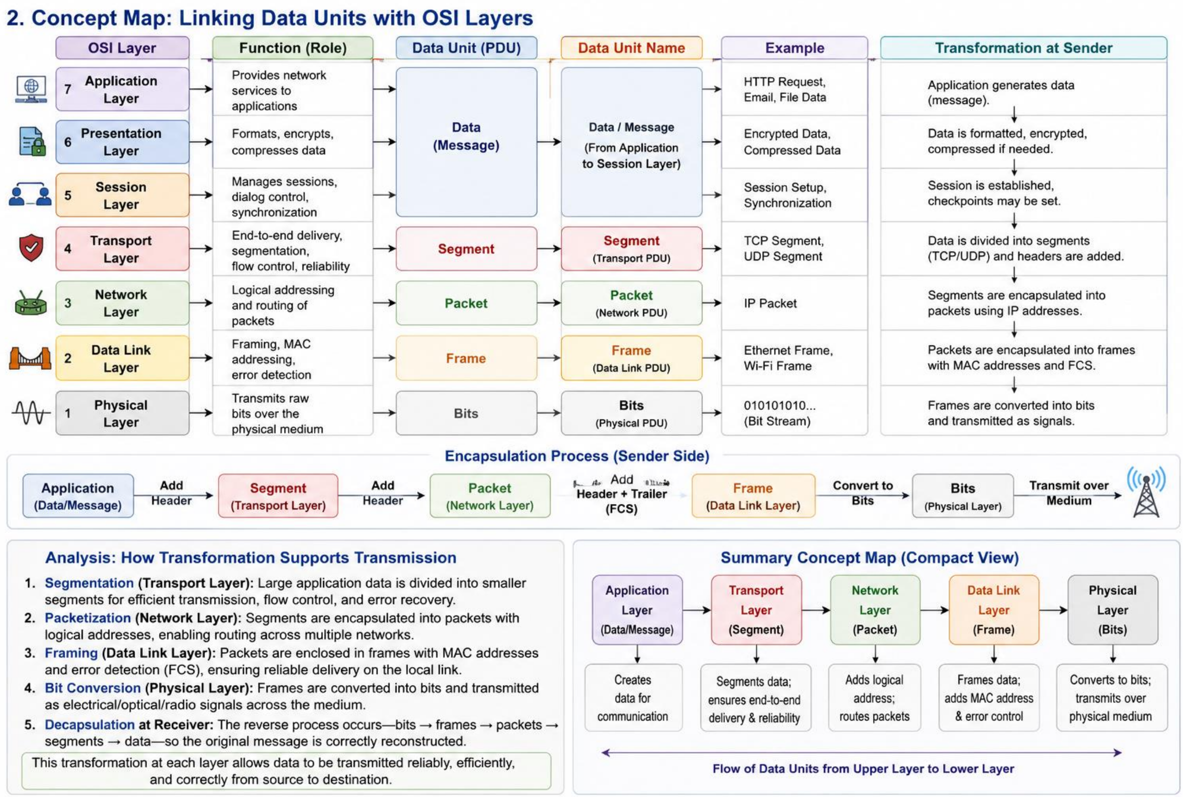
1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.



2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.



3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

The Data Link Layer uses MAC addresses to provide local delivery of data between devices on the same network. It encapsulates packets into frames, performs error detection, and ensures that each frame reaches the correct device within the local area network (LAN).

The Network Layer uses IP addresses to provide global delivery across different networks. It determines the best routing path and forwards packets through routers until they reach the destination network.

Together, these layers ensure complete data delivery by combining local communication (using MAC addresses) with end-to-end communication across multiple networks (using IP addresses). The Network Layer routes packets to the destination network, and the Data Link Layer delivers them to the correct device within that network, enabling reliable communication from source to destination.

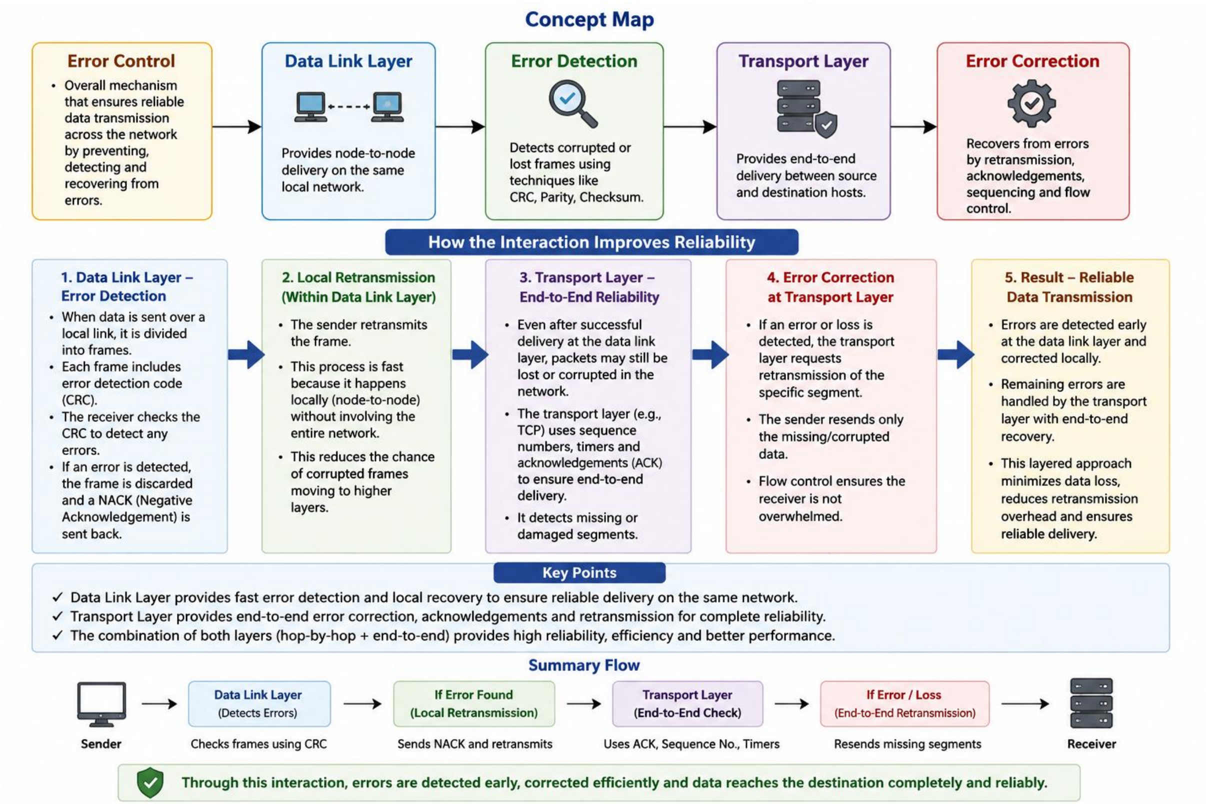
4. A concept map shows:

ErrorControl

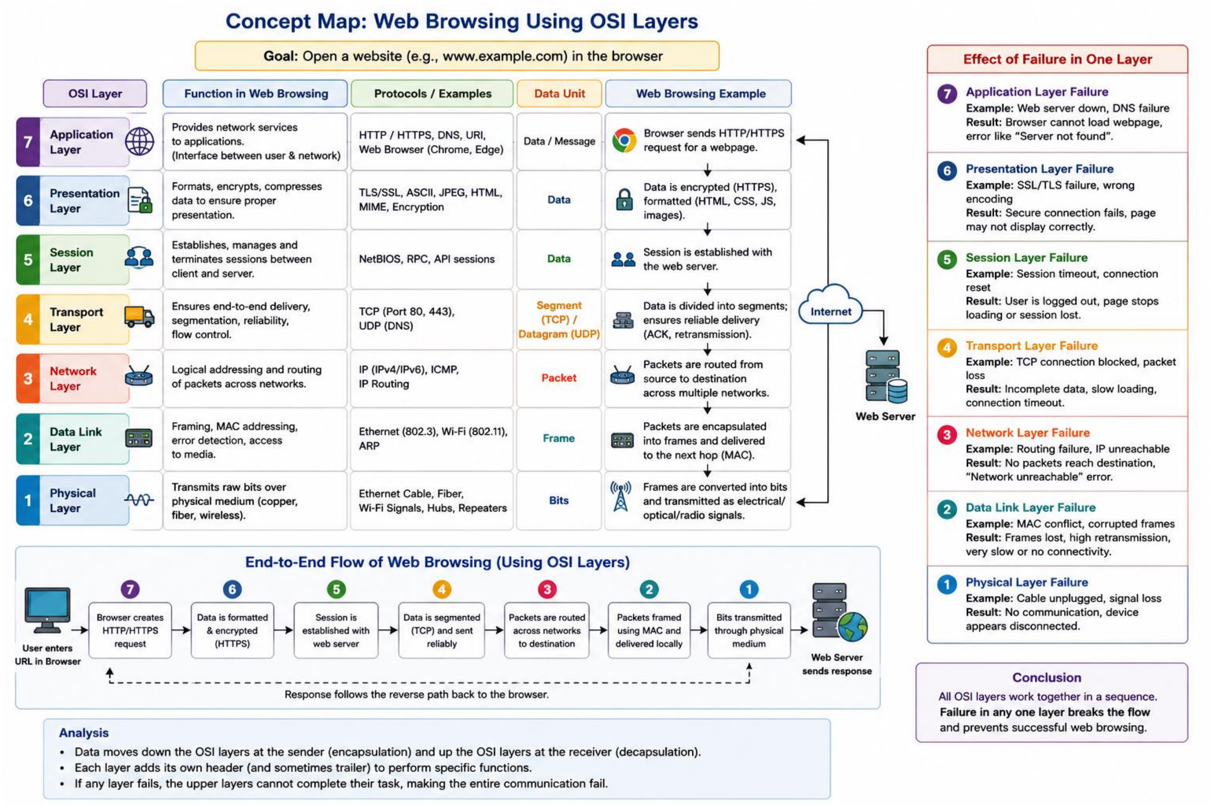
→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.



5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.



6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

The concept map indicates that the **Physical Layer** and **Data Link Layer** are functioning correctly, meaning the physical connection and local frame transmission are working properly. However, the **Network Layer has failed**, so packets cannot be routed to the destination network. As a result, the **Transport Layer** cannot establish end-to-end communication, and the **Application Layer** is unable to provide services such as web browsing, email, or file transfer.

The concept map helps in troubleshooting by showing exactly where communication stops. Since the Physical and Data Link layers are working, the fault is isolated to the **Network Layer**, allowing network administrators to focus on issues such as IP addressing, routing, or router configuration instead of checking the entire OSI model. This systematic approach reduces troubleshooting time and helps restore communication more efficiently.